

Every rule change. A traceable test story.

Rule2Test turns an insurance rule revision into reviewed regression tests and execution evidence you can hand to an auditor — with the AI never deciding pass or fail.

16.67% → 100.0%

designed boundary coverage, before to after

6/6

quality-gate contract cases matched

286 passing

automated tests on this build

Every figure in this deck is copied from an artifact in the repository. Synthetic data; not production measurements.

THE PROBLEM

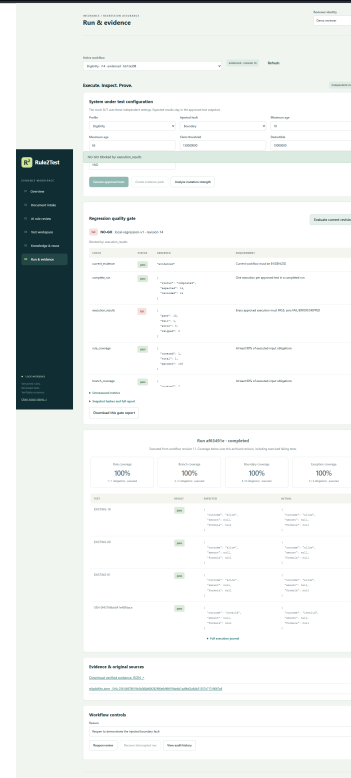
A rule changes. Nobody can prove what it broke.

- The boundary that moved is exactly the case the existing suite never covered.
- Nobody can say which tests the change invalidated, or who accepted the new ones.
- A green suite is presented as evidence, but nothing links a result back to the clause that caused it.
- Reviewers are asked to trust an AI summary of a document they cannot re-check line by line.

The gap is not test generation. It is **traceability and authority**.

A passing suite was never the point.

We inject an off-by-one at the exact threshold that moved. The generated boundary test catches it, and the release gate flips to **NO-GO** with a concrete counterexample — an age, not a stack trace.



Quality gate NO-GO: execution_results fails with 13 pass and 1 fail on the moved boundary.

HOW IT WORKS

Five stages, one audit trail.

Stage	What happens	Who decides
Ingest	Two rule versions; source bytes archived and hashed	Deterministic
Analyze	Rule delta, impact, bounded gap search, candidate tests	Deterministic
Review	Explicit, revision-bound approval per test	Human
Execute	Approved tests only, against an independent system under test	Deterministic
Prove	Evidence pack plus a GO / NO-GO regression gate	Deterministic

Each stage binds to a workflow revision. A stale action fails instead of silently overwriting work.

AI proposes. Engines decide. A human approves.

- **AI never computes expected.** A typed oracle does, from the reviewed rules.
- **AI never sees the answer.** Ground truth is compared only after the model returns.
- **Every extracted rule cites an exact source line.** If the quote is not literally in the document, we reject the output.
- **No retry, no repair, no fallback to mock.** A provider failure is classified and surfaced, never hidden.
- **Nothing is approved automatically** — not rules, not tests, not the AI's own proposals.

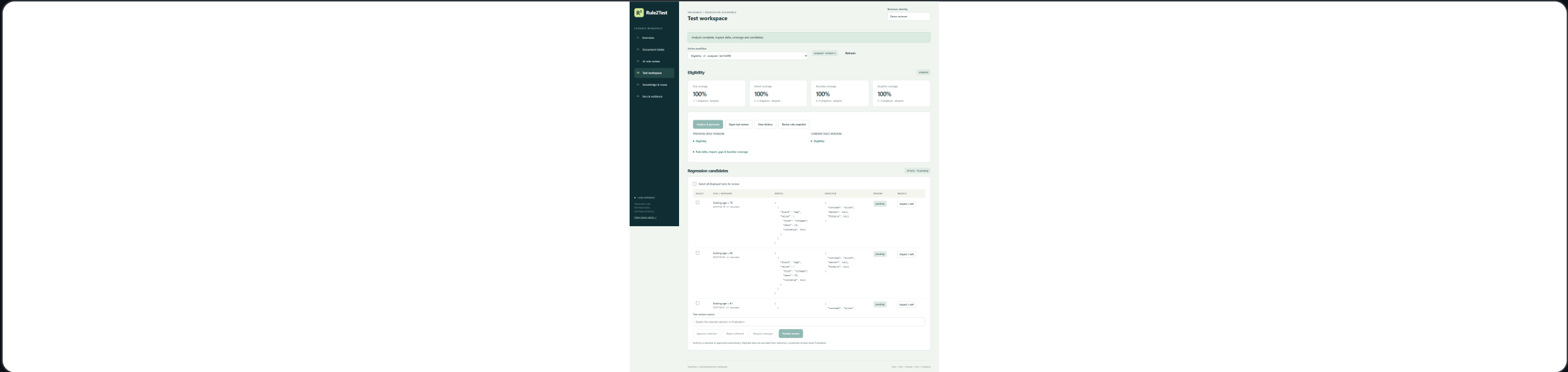
Find the cases nobody wrote.

16.67% → 100.0%

designed boundary coverage, baseline to final

100.0%

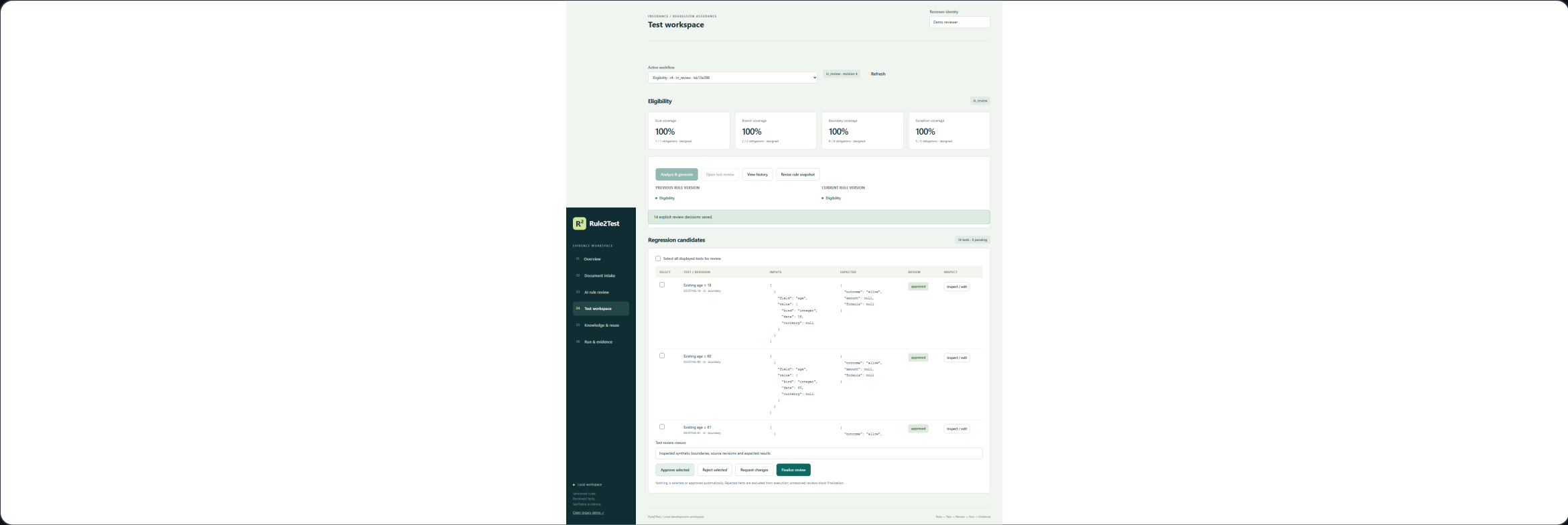
mutation score on the same synthetic scenario



Rule delta, impact, uncovered obligations and baseline coverage for the eligibility change.

Designed coverage on one synthetic scenario. Coverage is not a pass rate and not release approval.

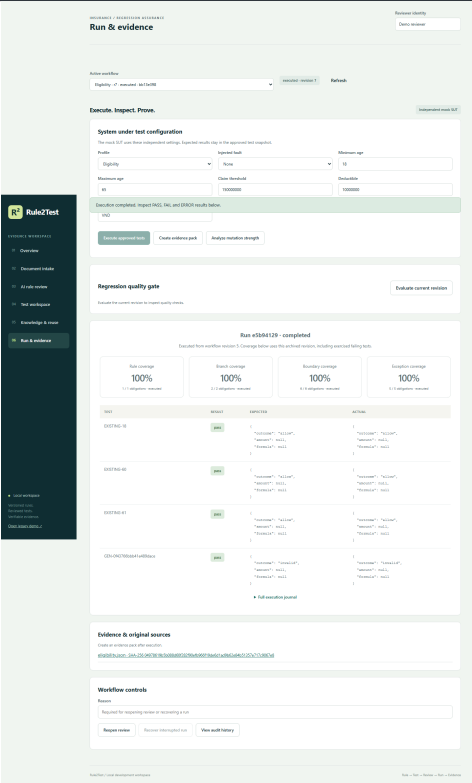
Approval is a person, a reason, and a revision.



Explicit selection, a written reason, and a per-test decision bound to the test revision.

Edit a test after approval and the approval is void. Reopening review clears approvals and invalidates the current GO context.

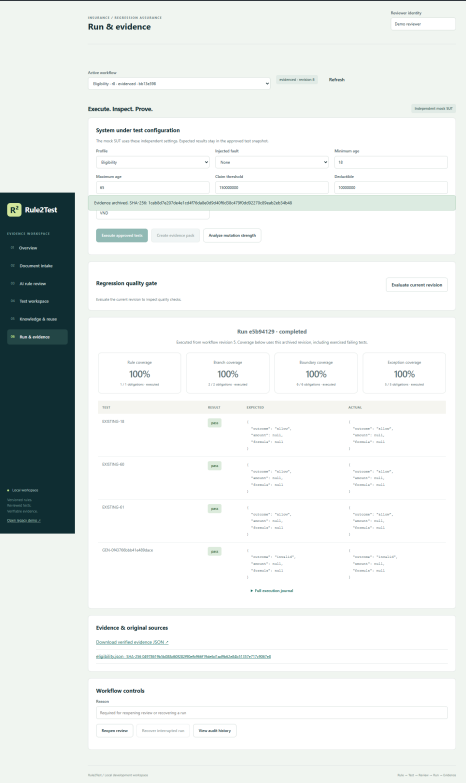
Expected and actual never share code.



Every row shows the oracle's expected value beside the independent system's actual value.

The system under test receives inputs only — never the expected result and never the rule snapshot.

Evidence an auditor can re-verify.



Evidence pack binding rules, approvals, inputs, expected, actual and the reviewer under one fingerprint.

SHA-256 detects corruption. It is an integrity check, not a digital signature, and it does not stop an administrator who rewrites both the payload and the hash.

MEASURED, HONESTLY

What the AI actually did.

4/10

live model, extraction cases matched,
prompt v2

5/10

schema and citation valid

9/9

retrieval top-1 over the reviewed corpus

4678 ms

median extraction latency

Run	Extraction	Meaning
Offline mock replay	5/10	Fixture replay only — not comprehension
Live model, prompt v1	2/10	First real-model run on ten authored Japanese cases
Live model, prompt v2	4/10	After inspecting v1 failures — not held-out validation

Ten authored cases, no SME-reviewed labels. Successful extraction is not reliable insurance interpretation.

Built to be debugged on stage.

286 passing

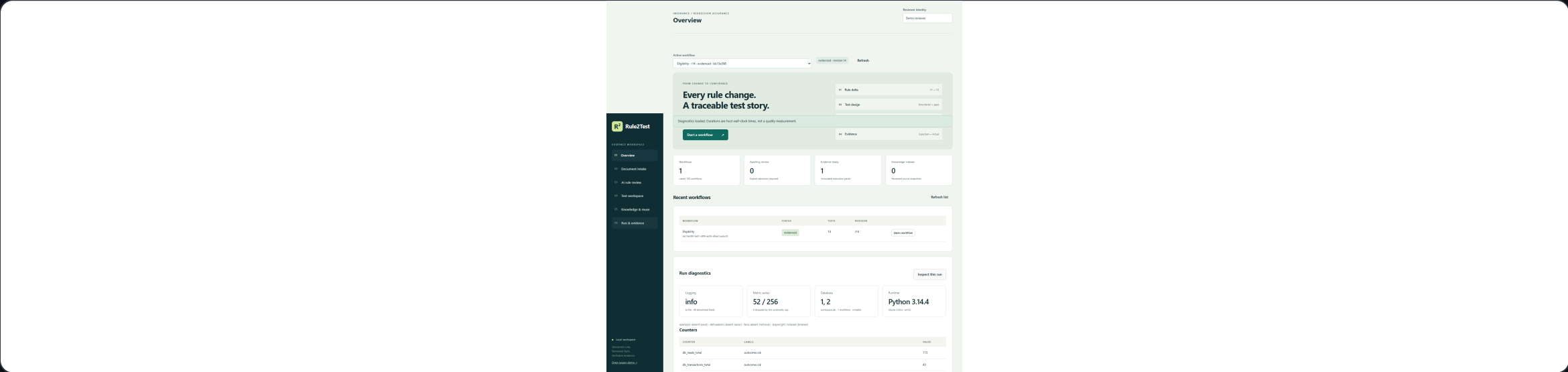
automated tests, standard library core

11

provider failure kinds, each with a remediation hint

0

automatic retries, fallbacks or auto-approvals



Run diagnostics: logging configuration, bounded metric series, database state and counters.

Logs use a field allowlist, so document text, prompts, reviewer names and session tokens cannot reach a log line. Counters are process-local and reset on restart.

What we do not claim.

- Not a general insurance reasoning engine — the supported rule DSL is deliberately small and rejects what it cannot express.
- Not production-ready: reviewer names are self-declared; there is no authentication, RBAC or deployment hardening.
- Not a release authorization — GO is a local regression assessment for one workflow revision.
- Not measured: reviewer acceptance, effort savings, production accuracy, token cost, SME-reviewed labels.
- Synthetic results throughout. Mock providers are labelled simulated everywhere they appear.

We would rather be trusted on five numbers than doubted on fifty.

NEXT

From a demo to a pipeline.

- Evaluate real extraction and embedding models against a broader, SME-labelled corpus.
- Richer decision-table analysis and metamorphic tests beyond single-field boundaries.
- Integrate a real system under test through the existing HTTP adapter.
- Authenticated roles, signed evidence, append-only audit storage and operational monitoring.

Every rule change becomes test coverage and auditable evidence.

Sources: docs/BENCHMARKS.md · docs/DIAGRAMS.md · docs/OBSERVABILITY.md · docs/COMPETITION_DEMO.md